Numerical Weather Prediction (NWP)

Individual Project

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Project Overview

Primary Goal

Develop a simplified but physically consistent mathematical model for NWP, focusing on **model formulation, numerical discretization, and validation** to address the core challenge of solving coupled nonlinear PDEs (e.g., Navier-Stokes) for atmospheric prediction.

Key Modeling Questions:

- How to simplify complex atmospheric PDEs while retaining critical physical dynamics?
- Which numerical schemes balance stability, accuracy, and computational feasibility?
- How to validate the model against real-world observations or benchmark data?

Scope: Focus on mesoscale short-range forecasts (0-72 hours) with hydrostatic approximation.

What is NWP?

- **Definition:** Computer-based numerical simulation and prediction of atmospheric and oceanic systems
- One of the major scientific achievements of the past 50 years
- Core component of operational weather forecasting
- Enables accurate forecasts exceeding one week
- **Key Dependencies:**
  - Higher model resolution (spatial/temporal discretization)
  - Quality numerical schemes (stable PDE solvers)
  - Improved physical parameterizations (sub-grid processes)
  - Satellite and observational data (validation/initialization)
  - Refined initial conditions (data assimilation)

Mathematical Modeling Framework

Core Foundation: PDE System Derived from Physics

NWP models are built on coupled nonlinear PDEs derived from fundamental conservation laws. This project focuses on a simplified but complete subset for tractability:

- Newton's laws of motion $\rightarrow$ Navier-Stokes equations (momentum)
- Conservation of energy $\rightarrow$ First law of thermodynamics
- Conservation of mass $\rightarrow$ Continuity equation
- Conservation of water vapor $\rightarrow$ Moisture equation
- Thermodynamic relationships $\rightarrow$ Equation of state + Hydrostatic balance

State Variables: $\boldsymbol{V} = (u, v, w)$, T, p, ρ, q (velocity, temperature, pressure, density, humidity)

Core Equation: Navier-Stokes Equations (Momentum)

Modeled Equation (Vector Form)

$$\frac{\partial \boldsymbol{V}}{\partial t} + \boldsymbol{V} \cdot \nabla \boldsymbol{V} + 2\boldsymbol{\Omega} \times \boldsymbol{V} + \frac{1}{\rho}\nabla p = \boldsymbol{F} + \boldsymbol{g}$$

Modeling Simplifications:

- Neglect vertical velocity w for mesoscale large-scale motions (hydrostatic approximation)
- Simplify friction $\boldsymbol{F}$ to linear drag in boundary layer (reduces nonlinearity)
- Fix Coriolis parameter $\boldsymbol{\Omega}$ (latitude-dependent constant for regional models)

Physical Interpretation (Retained Dynamics):

- $\frac{\partial \boldsymbol{V}}{\partial t}$: Local acceleration
- $\boldsymbol{V} \cdot \nabla \boldsymbol{V}$: Nonlinear advection (critical for weather systems)
- $2\boldsymbol{\Omega} \times \boldsymbol{V}$: Coriolis effect (Earth rotation)
- $\frac{1}{\rho}\nabla p$: Pressure gradient force (driving motion)

Key Supporting Equations (1/3)

Equation of State (Ideal Gas Law)

$$p = R\rho T(1 + 0.61q)$$

Moisture correction added for realism

Mass Conservation

$$\frac{d\rho}{dt} + \rho\nabla \cdot \boldsymbol{V} = 0$$

Ensures no mass creation/destruction, which is a core physical constraint for the PDE system

Key Supporting Equations (2/3)

Energy Conservation

$$c_v \frac{dT}{dt} + RT\nabla \cdot \boldsymbol{V} = Q$$

Q = radiative heating (parameterized as constant for model tractability)

Modeling Rationale

These three equations form the thermodynamic closure of the NWP model, linking pressure, temperature, density, and velocity to ensure physical consistency in numerical simulations.

Key Supporting Equations (3/3)

Hydrostatic Balance

$$\frac{\partial p}{\partial z} + g\rho = 0$$

Closes the vertical momentum equation by balancing vertical pressure gradient and gravity

Valid for large-scale mesoscale motions (simplifies full Navier-Stokes system)

Moisture Conservation (Simplified)

$$\frac{dq}{dt} = E - C$$

E = evaporation (constant boundary condition)

C = condensation (threshold-based, triggered when $q > q_{\text{sat}}$)

Numerical Methods: Solving the PDE System

Modeling Challenge

Continuous PDEs must be discretized into algebraic systems. Stability (CFL criterion) and accuracy are critical for meaningful forecasts.

Selected Numerical Schemes:

1. **Time Integration:** Leapfrog scheme
   - Explicit, second-order accurate: $\frac{Q_m^{n+1} - Q_m^{n-1}}{2\Delta t} = F(Q_m^n)$
   - Complies with CFL criterion: $\frac{c\Delta t}{\Delta x} \leq 1$ (ensures stability)
   - Computational cost feasible for simplified model
2. **Spatial Discretization:** Finite Difference Method
   - Centered differences for advection: $\frac{\partial Q}{\partial x} \approx \frac{Q_{m+1}^n - Q_{m-1}^n}{2\Delta x}$
   - Easy implementation for regional grid (Cartesian coordinates)
   - Balances accuracy and computational efficiency

Initial & Boundary Conditions (Model Setup)

Initial Value Problem (Critical for PDE Solution)

NWP models require well-posed initial conditions ($t = 0$) and boundary conditions to avoid unphysical solutions.

Modeled Initial Conditions:

- Use reanalysis data (e.g., ERA5) for temperature, pressure, and wind fields
- Simplify: Assume uniform humidity q (reduces initialization complexity)
- Balance check: Ensure hydrostatic equilibrium ($\frac{\partial p}{\partial z} = -g\rho$) to avoid spurious oscillations

Boundary Conditions:

- Open boundaries: Radiation conditions (waves exit without reflection)
- Surface boundary: Fixed temperature (land surface) or heat flux (ocean)

Data Assimilation: Optimizing Initial Conditions

3D-Var Cost Function (Model Calibration)

To improve initial conditions, we minimize the discrepancy between model background and observations:

$$J = \frac{1}{2}(\boldsymbol{X} - \boldsymbol{X}_B)^T \boldsymbol{B}^{-1}(\boldsymbol{X} - \boldsymbol{X}_B) + \frac{1}{2}(\boldsymbol{Y} - H\boldsymbol{X})^T \boldsymbol{R}^{-1}(\boldsymbol{Y} - H\boldsymbol{X})$$

Modeling Simplification:

- Assume diagonal error covariance matrices $\boldsymbol{B}$ and $\boldsymbol{R}$ (reduces computational cost)
- Linear observation operator H (direct mapping from model state to observations)
- Solve minimization using conjugate gradient method (no need for full matrix inversion)

Goal: Generate optimal initial state $\boldsymbol{X}$ that balances model dynamics and observational data.

Project Milestones

Phase 1: Model Formulation

- Finalize simplified PDE system
- Define model domain
- Set discretization parameters ($\Delta x = 10km$, $\Delta t = 15min$)

Phase 2: Numerical Implementation

- Code finite difference discretization for PDEs
- Implement leapfrog time-stepping scheme
- Add 3D-Var data assimilation

Phase 3: Validation & Analysis

- Validate model against ERA5 reanalysis data
- Assess stability (CFL compliance) and forecast skill (ACC $> 60\%$)
- Analyze impact of numerical schemes on results

Validation Plan: Measuring Model Performance

Key Metrics (Standard for NWP Modeling)

1. **Anomaly Correlation Coefficient (ACC)**
   - Measures agreement between forecast and observed anomalies
   - Target: ACC $> 60\%$ for 24-hour forecasts (indicates skill)
2. **Root Mean Square Error (RMSE)**
   - Quantifies magnitude of errors in temperature/wind fields
   - Target: RMSE $< 2°\mathrm{C}$ (temperature) and $< 2\,\mathrm{m/s}$ (wind speed)
3. **Stability Check**
   - Verify no exponential growth of errors (CFL criterion satisfaction)
   - Ensure model runs for 72 hours without numerical blowup

Benchmark Data: ERA5 reanalysis (ECMWF) – gold standard for NWP validation.

Challenges & Mitigations

Key Modeling Challenges

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Applications & Impact of the Model

Academic Impact:

- Demonstrates end-to-end mathematical modeling of a complex PDE-based system
- Provides a simplified framework for understanding NWP numerical challenges

Practical Relevance:

- Short-range mesoscale forecasts (e.g., local storms, temperature extremes)
- Foundation for extended models (e.g., adding ocean-sea ice coupling)
- Educational tool for numerical PDE solving in geosciences

Limitations (Future Work):

- Extend to non-hydrostatic formulation for convective storms
- Add more complex physical parameterizations (clouds, radiation)
- Scale to global model with spectral methods

Summary

1. **Objective:** Develop a simplified but physically consistent NWP model, focusing on PDE solution, numerical stability, and validation.
2. **Methods:**
   - Simplified PDE system (Navier-Stokes + supporting equations)
   - Finite difference discretization + leapfrog time-stepping
   - 3D-Var data assimilation for initial conditions
   - Validation against ERA5 reanalysis
3. **Milestones:** 3-phase timeline (formulation $\rightarrow$ implementation $\rightarrow$ validation)
4. **Value:** Addresses core challenge of solving Navier-Stokes for weather prediction, with clear metrics for success.